Sample Size Calculator (R Shiny) — Scope Review Checklist

For colleague review: study designs, statistical tests, and planned scope

This document lists the study designs (Part 1) and statistical tests (Part 2) planned for an R Shiny sample-size/power calculator, with the key inputs and R package each module will use. **CORE** items (orange) are planned for the first release; **ADV** items are later-phase/advanced additions. Please review for completeness, priority, and clinical/methodological relevance to your area, and use the **Reviewer remarks** column to note anything you'd add, drop, reprioritize, or would use yourself.

Part 1 — By Study Design

1. Descriptive / Estimation studies

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Single proportion / prevalence	CORE	expected p, precision, confidence, N (optional)	epiR::epi.sssimpleestb; presize::prec_prop	
Single mean	CORE	SD, precision (CI half-width), confidence	presize::prec_mean	
Incidence rate estimation	CORE	expected rate, person-time, precision	presize::prec_rate	
Ratio estimation (RR/OR/rate ratio precision)	ADV	expected ratio, precision	presize::prec_riskratio/prec_or/prec_rateratio	
Precision / CI-width engine	CORE	generic engine feeding estimation modules	custom (Rothman-Greenland)	
Finite population correction	CORE	population size N, computed n	formula: n/(1+n/N)	
Variance / SD estimation precision	ADV	expected variance, precision	custom	

2. Cross-sectional / Survey designs

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Simple random sample, proportion	CORE	as single proportion, survey framing	as above	
Stratified sampling	CORE	strata sizes, within-stratum variance, allocation	custom	
Cluster / multistage with design effect	CORE	ICC (rho), avg cluster size m	DEFF = 1+(m-1)rho	
Complex/multistage survey (PPS)	ADV	rho, take b	custom (DHS WP30)	
Lot Quality Assurance Sampling (LQAS)	ADV	thresholds, decision rule, alpha/beta	simulation	
Two-stage cluster survey (WHO EPI 30x7/30x30)	ADV	fixed design parameters	reference formulas	
Capture-recapture	ADV	Chapman estimator inputs	Rcapture (analysis)	

3. Analytical observational studies

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Cohort / cross-sectional (RR, risk diff.)	CORE	p0, RR or p1, ratio unexposed:exposed	epiR::epi.sscohortc / epi.sscohortt	
Unmatched case-control	CORE	% controls exposed, OR, controls:cases ratio	epiR::epi.sscc	
Matched case-control 1:1 and 1:m	CORE	exposure prevalence in controls, OR, m controls/case	epiR::epi.sscc(method='matched')	
Frequency-matched design	ADV	matching factor distribution	custom	
Nested case-control / case-cohort	ADV	sub-cohort sampling fraction	custom (Prentice)	
Case-crossover	ADV	exposure discordance	custom	
Self-controlled case series	ADV	event/exposure windows	SCCS	
Ecological studies	ADV	aggregate-level exposure/outcome	custom, note caveats	
Comparison of two incidence rates	CORE	two person-time rates	epiR::epi.sscompc	

4. Diagnostic & screening studies

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Sensitivity / specificity given prevalence	CORE	expected Se or Sp, prevalence, precision	presize::prec_sens / prec_spec	
PPV / NPV	CORE	Se, Sp, prevalence, precision	presize (derived)	
Likelihood ratios (LR+, LR-)	ADV	Se, Sp, precision	presize::prec_lr	
Single ROC AUC vs null (0.5)	CORE	expected AUC, allocation ratio, alpha, power	pROC::power.roc.test	
Compare two AUCs (paired/unpaired)	ADV	AUC1, AUC2, correlation, allocation	pROC::power.roc.test	
Diagnostic odds ratio	ADV	Se, Sp derived	custom	
Multi-reader multi-case (MRMC)	ADV	reader/case variance components	simulation / iMRMC	
Index-test vs reference-standard accuracy	ADV	accuracy target, prevalence	custom	

5. Agreement / reliability studies

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Cohen's kappa	CORE	expected kappa, null kappa, category proportions, raters	kappaSize	
Weighted / Fleiss kappa (>2 raters)	ADV	as above, multiple raters	kappaSize	
Intraclass correlation coefficient (ICC)	CORE	expected ICC, #raters/measurements, CI/power	ICC.Sample.Size; presize::prec_icc	
Bland-Altman limits of agreement	ADV	SD of differences, CI width	presize::prec_lim_agree	
Lin's concordance correlation coefficient	ADV	expected CCC, precision	custom	

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Cronbach's alpha	ADV	expected alpha, #items	presize::prec_cronb	
Test-retest reliability	ADV	ICC/kappa over time	custom	

6. Clinical trials

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Parallel 2-arm superiority (means/proportions)	CORE	effect size, alpha, power	stats::power.t.test; power.prop.test; pwr	
Multi-arm superiority (multiplicity)	ADV	#arms, effect sizes, alpha adjustment	custom	
Non-inferiority (means/proportions)	CORE	NI margin, expected difference, one-sided alpha	SampleSize4ClinicalTrials; TrialSize	
Equivalence (TOST)	CORE	equivalence bounds, expected difference	PowerTOST; TOSTER	
Bioequivalence crossover (2x2 & replicate)	ADV	within-subject CV, GMR, BE limits 0.80-1.25	PowerTOST::sampleN.TOST	
Cluster-randomized trials	ADV	ICC/DEFF, cluster size variability	clusterPower; CRTSize	
Stepped-wedge CRT	ADV	steps, clusters/step, ICC	swCRTdesign; SWSamp; clusterPower	
Factorial designs	ADV	main effects, interaction	custom	
Crossover trials (2x2, continuous)	CORE	within-subject variance, period effect	PowerTOST	
N-of-1 trials	ADV	cycles, within-subject variance	custom	
Pilot / feasibility (rules of thumb)	CORE	none / expected effect	rule: n=12/arm (Julious 2005)	
Dose-finding / phase I (3+3, CRM)	ADV	dose levels, toxicity target	dfcrm; bcrm	
Phase II single-arm (Simon two-stage)	ADV	p0, p1, alpha, beta	clinfun::ph2simon; ph2mult; gen2stage	
Group sequential & adaptive designs	ADV	#looks, spending function	gsDesign; rpact	
Blinded/unblinded SSR	ADV	interim variance/effect re-estimate	gsDesign::ssrCP; spass	
Platform / basket / umbrella trials	ADV	arm entry/exit rules	simulation	

7. Survival / time-to-event studies

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Log-rank test	CORE	HR or median survivals, accrual+follow-up, allocation	powerSurvEpi::ssizeCT; gsDesign::nSurv	
Cox proportional hazards	CORE	HR, covariate type (binary/non-binary)	powerSurvEpi::ssizeEpi / powerEpi	
Exponential survival (parametric)	ADV	hazard rates, accrual/follow-up	gsDesign::nSurv	
Accrual + follow-up (non-uniform entry)	ADV	accrual pattern	gsDesign::nSurv; npsurvSS	
Competing risks	ADV	subdistribution hazard	powerCompRisk	

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Restricted mean survival time (RMST)	ADV	time horizon, expected RMST diff.	SSRMST	
Non-proportional hazards / weighted log-rank	ADV	weighting scheme	custom (Yung & Liu)	
One-sample survival vs historical control	ADV	historical survival curve	custom	

8. Prediction / prognostic model studies

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Model development (binary/continuous/TTE)	CORE	#candidate predictors, anticipated R2/C-stat, outcome prevalence	pmsampsize	
External validation - binary outcome	ADV	expected performance, prevalence	pmvalsampsize	
External validation - time-to-event	ADV	expected performance, event rate	custom (Riley 2022)	
Model updating / recalibration / ML sample size	ADV	recalibration scope	custom	

9. Genetic / omics & high-dimensional

Item	Pri.	Key inputs	R package / method	Reviewer remarks
GWAS / candidate-gene association	ADV	MAF, OR/GRR, prevalence, genome-wide alpha	genpwr; Quanto; Haplin	
Gene-environment interaction	ADV	interaction OR, MAF	genpwr; Quanto	
Microarray/RNA-seq differential expression (FDR)	ADV	fold-change, FDR	ssize; RNASeqPower; PROPER	
Multiple-testing sample-size inflation	ADV	#tests, FDR/Bonferroni	custom	

10. Psychometric / questionnaire / scale development

Item	Pri.	Key inputs	R package / method	Reviewer remarks
EFA/CFA subjects-per-item rules of thumb	ADV	#items, ratio rule	custom	
CFA/SEM via RMSEA	ADV	df, null vs alt RMSEA, alpha, power	semPower; WebPower::wp.sem.rmsea; semTools	
IRT calibration; validation studies	ADV	item bank size, model	custom	

11. Qualitative & mixed methods (guidance only)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Saturation-based guidance	ADV	narrative guidance panel	n/a - Guest et al. 2006	
Information power	ADV	narrative guidance panel	n/a - Malterud et al. 2016	
Purposive/theoretical sampling guidance	ADV	narrative guidance panel, no numeric N	n/a	

12. Health economics / outcomes

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Cost-effectiveness study sample size	ADV	net-benefit framework inputs	custom	
Value of information (EVPI/EVSI)	ADV	decision model parameters	voi	

13. Pharmacovigilance / surveillance

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Rule of three	CORE	n (sample size) or target incidence	3/n rule - Hanley & Lippman-Hand	
Detecting rare adverse events / signal thresholds	ADV	expected incidence, detection probability	binomial/Poisson custom	

14. Laboratory / animal / preclinical

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Resource equation method (E 10-20)	ADV	#groups, #animals	formula (Mead; Charan & Kantharia)	
Animal-study rank tests	ADV	Lehmann alternative	clinfun	
Method-comparison studies (Deming/Passing-Bablok)	ADV	expected agreement/LoA	presize (LoA)	

15. Meta-analysis

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Trial sequential analysis / optimal information size	ADV	control event rate, RRR, alpha, beta, heterogeneity D2	RTSA	
Power / effective N for (network) meta-analysis	ADV	#studies, heterogeneity	metapower	

16. Real-world evidence / advanced observational

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Propensity-score-matched designs	ADV	matching efficiency factor	custom	
RWE database studies (precision-driven)	ADV	large-N precision target	custom	
Subgroup & interaction testing ('4x rule')	ADV	main effect size, interaction size	custom (Brookes et al. 2004)	

17. Repeated measures / longitudinal designs

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Linear mixed models, rate of change	ADV	visit times, variance components, slope diff.	longpower::lmmpower	
MMRM with missing data	ADV	correlation/AR(1), retention, SD, allocation	longpower::power.mmrm / power.mmrm.ar1	
Multilevel (2-/3-level, with dropout)	ADV	time points, subjects/cluster, ICCs, dropout	powerlmm	
GEE-based (marginal models)	ADV	correlation structure	longpower::liu.liang.linear.power	
Time-varying exposure	ADV	exposure ICC, correlation structure	OPTITXS (non-CRAN)	
Growth curve models	ADV	simulation-based	simr	

Part 2 — By Statistical Test (mirrors G*Power's catalog)

t tests (G*Power t family)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
One-sample t (mean vs constant)	CORE	effect size d, alpha, power	pwr::pwr.t.test(type='one.sample')	
Two independent means, equal n	CORE	d, alpha, power	pwr::pwr.t.test(type='two.sample')	
Two independent means, unequal n	CORE	d, n ratio	pwr::pwr.t2n.test	
Welch t (unequal variances)	ADV	SD1, SD2, ratio	MESS::power_t_test / simulation	
Paired t	CORE	dz (mean & SD of differences)	pwr::pwr.t.test(type='paired')	
Point-biserial correlation	ADV	r, alpha, power	custom	
Regression coefficient / bivariate slope	ADV	slope, SDx, SDy	custom	

z tests (G*Power z family)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Single proportion	CORE	effect size h (or p vs p0)	pwr::pwr.p.test	
Two proportions	CORE	h, alpha, power, ratio	pwr::pwr.2p.test; pwr.2p2n.test	
Logistic regression (simple/multiple)	ADV	OR, predictor distribution, R2 with covariates	WebPower::wp.logistic	
Poisson regression	ADV	rate ratio, exposure distribution	WebPower::wp.poisson	

F tests (G*Power F family)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
One-way ANOVA	CORE	f, k groups, alpha, power	pwr::pwr.anova.test; WebPower::wp.anova	
Two-way / factorial ANOVA	ADV	f, numerator df	custom	
Repeated-measures ANOVA	ADV	f, #measurements, correlation, nonsphericity	WebPower::wp.rmanova	
ANCOVA	ADV	f, #covariates, #groups	WebPower family / nMaster	
MANOVA (incl. Hotelling's T2)	ADV	multivariate effect size	custom	
Mixed (split-plot) ANOVA	ADV	between/within effects	custom	
Multiple linear regression, R2 vs zero	CORE	effect size f2, #predictors	pwr::pwr.f2.test	
Multiple regression, R2 increase (hierarchical)	ADV	f2 for added predictor set	custom	
Random-model multiple regression	ADV	as above, random-effects model	custom	

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Planned contrasts / post-hoc with multiplicity	ADV	contrast effect sizes	custom	

Chi-square tests (G*Power chi-sq family)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Goodness of fit	CORE	effect size w, df	pwr::pwr.chisq.test (ES.w1)	
Test of independence / association (R x C)	CORE	w, df	pwr::pwr.chisq.test (ES.w2)	
McNemar's (paired proportions)	ADV	OR, proportion discordant	custom	
Cochran's Q (k paired proportions)	ADV	#conditions, discordance	custom	
Mantel-Haenszel (stratified OR)	ADV	stratum-specific OR	custom	
Cochran-Armitage trend	ADV	dose-response proportions	custom	
Single-variance chi-square test	ADV	expected vs null variance	custom	

Exact tests (G*Power Exact family)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Fisher's exact (2x2)	CORE	p1, p2, allocation	custom	
Binomial test (single proportion)	CORE	p, p0	custom	
Exact Poisson (rare event vs standard)	ADV	expected vs standard rate	custom	
Correlation, bivariate normal (exact)	ADV	r, alpha, power	custom	
Proportions difference, unconditional exact	ADV	p1, p2	custom	

Correlation & regression

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Pearson correlation	CORE	r, alpha, power	pwr::pwr.r.test	
Spearman / Kendall's tau	ADV	ARE-inflated r	custom	
Partial correlation	ADV	r, #controlled variables	custom	
Compare two independent correlations	ADV	q (Fisher r-to-z)	custom	
Compare two dependent correlations	ADV	common/non-common index	custom	
Logistic / Poisson / negative binomial regression	ADV	see z-family entries	WebPower::wp.logistic / wp.poisson	

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Cox regression	ADV	HR, event proportion, covariate SD/corr.	powerSurvEpi	
Ordinal (proportional odds) / multinomial regression	ADV	cumulative-odds effect	custom (Whitehead 1993)	

Non-parametric tests (ARE inflation / simulation)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Mann-Whitney U / Wilcoxon rank-sum	ADV	d x ARE (approx 1.05 for normal)	custom / nMaster	
Wilcoxon signed-rank	ADV	parent distribution assumption	custom / nMaster	
Kruskal-Wallis	ADV	ANOVA n x ARE	custom / simulation	
Friedman	ADV	RM-ANOVA basis	custom / simulation	
Sign test	ADV	exact binomial basis	pwr / nMaster	
Jonckheere-Terpstra (ordered alternatives)	ADV	odds ratio per group	clinfun	
Kolmogorov-Smirnov; runs test; permutation	ADV	distribution-free	simulation (simr / custom)	

Survival tests

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Log-rank	CORE	HR, accrual/follow-up	powerSurvEpi; gsDesign::nSurv	
Cox PH	ADV	HR, covariate	powerSurvEpi	
Gray's test (competing risks)	ADV	subdistribution hazard	powerCompRisk	
RMST comparison	ADV	time horizon	SSRMST	
Weighted log-rank (non-PH)	ADV	weighting scheme	npsurvSS / simulation	

Agreement statistics

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Kappa	CORE	kappa0, kappa1, proportions, raters	kappaSize	
ICC (forms 1,1 / 2,1 / 3,1)	CORE	ICC, #raters/measurements, CI/power	ICC.Sample.Size; presize::prec_icc	
CCC; Bland-Altman LoA	ADV	expected agreement / SD of differences	presize	

Multiple-comparison procedures

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Bonferroni-adjusted alpha	ADV	#comparisons m, base alpha	wrapper on any calculator	
Holm / FDR (Benjamini-Hochberg)	ADV	#tests, FDR level	ssize.fdr	

Equivalence / non-inferiority test forms

Item	Pri.	Key inputs	R package / method	Reviewer remarks
TOST for means	CORE	equivalence bounds, expected difference	PowerTOST; TOSTER::power_t_TOST	
TOST for proportions	CORE	equivalence bounds, expected difference	TOSTER; PowerTOST	
NI margins for RR/OR/HR/difference	ADV	margin, expected ratio/difference	TrialSize; SampleSize4ClinicalTrials	

Multilevel / mixed-model / GEE tests

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Mixed-model tests (2/3-level)	ADV	variance components, ICC	powerlmm; longpower	
GEE-based tests	ADV	correlation structure	longpower::liu.liang.linear.power	

Bayesian sample-size approaches

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Bayes factor design analysis	ADV	expected BF, stopping rule	BFDA; bfpwr	
Assurance / expected power	ADV	prior on effect size	PowerTOST (exp.*)	
Average coverage criterion; HPD width	ADV	target interval width	BayesianPower	

Simulation-based power (no closed form)

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Monte Carlo for GLMMs / complex designs	ADV	fitted model, simulation reps	simr; Superpower; SimEngine	

Sequential / adaptive test procedures

Item	Pri.	Key inputs	R package / method	Reviewer remarks
O'Brien-Fleming / Pocock boundaries	ADV	#looks, spending function	gsDesign; rpact	
Lan-DeMets alpha spending	ADV	spending function shape	gsDesign; rpact	
Conditional power; futility; SSR	ADV	interim effect estimate	gsDesign::ssrCP; condPower	

Tests for variance / rates / counts

Item	Pri.	Key inputs	R package / method	Reviewer remarks
Single-/two-variance (F) tests	ADV	expected variance ratio	custom	
One-/two-sample Poisson rate	ADV	expected rates, person-time	epiR	